

Associative Array, Multi-Dimensional Array, Function, Array Function, Array Sorting

- Md. Morshedul Arefin

Associative Array

Example - Normal:

```
$persons = [ "name" => "Smith", "age" => 30 ];  
echo $persons["name"]; // Smith  
echo $persons["age"]; // 30
```

Example - foreach loop:

```
$persons = [ "name" => "Smith", "age" => 30 ];  
foreach ($persons as $key => $value) {  
    echo "$key: $value\n";  
}
```

Multidimensional Array

Example - no loop:

```
$persons = [ ["Smith",30], ["Robin",20], ["Peter",25] ];  
echo $persons[0][0];  
echo $persons[0][1];  
echo $persons[1][0];  
echo $persons[1][1];  
echo $persons[2][0];  
echo $persons[2][1];
```

Multidimensional Array (Continue ...)

Example - for loop:

```
$persons = [ ["Smith",30], ["Robin",20], ["Peter",25] ];  
for($i=0;$i<count($persons);$i++) {  
    for($j=0;$j<count($persons[$i]);$j++) {  
        echo $persons[$i][$j];  
    }  
}
```

Multidimensional Array (Continue ...)

Example - foreach loop:

```
$persons = [ ["Smith",30], ["Robin",20], ["Peter",25] ];  
foreach($persons as $person) {  
    foreach ($person as $val) {  
        echo $val;  
    }  
}
```

Multidimensional Array (Continue ...)

Example - parent is indexed, child is associative

```
$persons = [  
    [ "name" => "Smith", "age" => 30 ],  
    [ "name" => "Robin", "age" => 20 ],  
    [ "name" => "Peter", "age" => 25 ]  
];  
foreach($persons as $person) {  
    foreach($person as $key => $value) {  
        echo "$key: $value";  
    }  
}
```

Multidimensional Array (Continue ...)

Example - parent and child both are associative

```
$persons = [  
    "person1" => [ "name" => "Smith", "age" => 30 ],  
    "person2" => [ "name" => "Robin", "age" => 20 ],  
    "person3" => [ "name" => "Peter", "age" => 25 ]  
];  
foreach($persons as $personKey=>$personVal) {  
    echo $personKey;  
    foreach($personVal as $key => $value) {  
        echo "$key: $value";  
    }  
}
```

Function

1. User-defined
2. Built-in

Reference:

<https://www.php.net/manual/en/funcref.php>

Function (Continue ...)

Example - No parameter - print into body:

```
function country() {  
    echo "Bangladesh";  
}  
country();
```

Example - No parameter - print outside:

```
function country() {  
    return "Bangladesh";  
}  
echo country();
```

Example - with Single Parameter - print into body:

```
function show($name) {  
    echo $name;  
}  
show("Smith");
```

Function (Continue ...)

Example - with Single Parameter - print outside:

```
function show($name) {  
    return $name;  
}  
echo show("Smith");
```

Example - with Multiple Parameters:

```
function add($a, $b) {  
    return ($a+$b);  
}  
echo add(10,20); // 30  
echo add(50,10); // 60  
echo add(100,200); // 300
```

Function (Continue ...)

Example - Default Parameter Values:

```
function add($a = 0, $b = 0, $c = 0) {  
    return ($a+$b+$c);  
}  
echo add(10,20); // 30
```

Example - Practical:

```
function pyramid($number) {  
    for($i=1;$i<=$number;$i++) {  
        for($j=1;$j<=$i;$j++) {  
            echo $j." ";  
        }  
        echo "\n";  
    }  
}  
pyramid(3); pyramid(7); pyramid(9);
```

Array Function

Reference:

<https://www.php.net/manual/en/ref.array.php>

Array Function (Continue ...)

Example - Common Functions:

```
$arr = [12, 33, 55, 24, 67];  
echo count($arr); // 5
```

```
$arr = [12, 33, 55, 24, 67];  
var_dump($arr);
```

```
$arr = [12, 33, 55, 24, 67];  
print_r($arr);
```

```
$arr = [12, 33, 55, 24, 67];  
var_dump(in_array(55,$arr)); // bool(true)
```

```
$arr = [33, 12, 55, 89, 24, 67];  
echo max($arr); // 89
```

```
$arr = [33, 12, 55, 89, 24, 67];  
echo min($arr); // 12
```

Array Function (Continue ...)

Example - Get Unique Values - Key not properly indexing:

```
$arr = [33, 12, 55, 33, 89, 12, 24, 33, 22, 55, 67];  
print_r(array_unique($arr));
```

Output:

Array

(

[0] => 33

[1] => 12

[2] => 55

[4] => 89

[6] => 24

[8] => 22

[10] => 67

)

Array Function (Continue ...)

Example - Get Unique Values - Key properly indexing:

```
$arr = [33, 12, 55, 33, 89, 12, 24, 33, 22, 55, 67];  
print_r(array_values(array_unique($arr)));
```

Output:

Array

(

[0] => 33

[1] => 12

[2] => 55

[3] => 89

[4] => 24

[5] => 22

[6] => 67

)

Array Function (Continue ...)

Example - String to Array - explode()

```
$date = "2023-04-25";  
print_r(explode("-", $date));
```

Output:

Array

```
(  
    [0] => 2023  
    [1] => 04  
    [2] => 25  
)
```

Array Function (Continue ...)

Example - Array to String - implode()

```
$cities = ["Khulna", "Dhaka", "Sylhet"];  
echo implode("-", $cities); // Khulna-Dhaka-Sylhet
```

Array Sort

Example - sort() - Indexed array - Normal Sort (small to large):

```
$arr = [33, 12, 55, 89, 24, 67];  
sort($arr);  
print_r($arr);
```

Output:

Array

```
(  
    [0] => 12  
    [1] => 24  
    [2] => 33  
    [3] => 55  
    [4] => 67  
    [5] => 89  
)
```

Array Sort (Continue ...)

Example - rsort() - Indexed array - Reverse Sort (large to small):

```
$arr = [33, 12, 55, 89, 24, 67];  
rsort($arr);  
print_r($arr);
```

Output:

Array

```
(  
    [0] => 89  
    [1] => 67  
    [2] => 55  
    [3] => 33  
    [4] => 24  
    [5] => 12  
)
```

Array Sort (Continue ...)

Example - asort() - Associative Array - Normal Sort (small to large) -

According to the value:

```
$marks = [ "Cooper" => 64, "Adam" => 80, "Bent" => 45 ];  
asort($marks);  
print_r($marks);
```

Output:

```
Array  
(  
    [Bent] => 45  
    [Cooper] => 64  
    [Adam] => 80  
)
```

Array Sort (Continue ...)

Example - ksort() - Associative Array - Normal Sort (small to large) -

According to the key:

```
$marks = [ "Cooper" => 64, "Adam" => 80, "Bent" => 45 ];  
ksort($marks);  
print_r($marks);
```

Output:

```
Array  
(  
    [Adam] => 80  
    [Bent] => 45  
    [Cooper] => 64  
)
```

Array Sort (Continue ...)

Example - arsort() - Associative Array - Reverse Sort (large to small) -

According to the value:

```
$marks = [ "Cooper" => 64, "Adam" => 80, "Bent" => 45 ];  
arsort($marks);  
print_r($marks);
```

Output:

```
Array  
(  
    [Adam] => 80  
    [Cooper] => 64  
    [Bent] => 45  
)
```

Array Sort (Continue ...)

Example - krsort() - Associative Array - Reverse Sort (large to small) -

According to the key:

```
$marks = [ "Cooper" => 64, "Adam" => 80, "Bent" => 45 ];  
krsort($marks);  
print_r($marks);
```

Output:

```
Array  
(  
    [Cooper] => 64  
    [Bent] => 45  
    [Adam] => 80  
)
```